

Time-Division Spectrum Sharing and Coordination Between Beam-Hopping NGSO Satellites and Terrestrial Networks

Kai Chang¹, Graduate Student Member, IEEE, Jianxiu Wang²,
Bingkun Liu³, Member, IEEE, and Linling Kuang⁴, Senior Member, IEEE

Abstract—Due to the limited frequency resources, the future integrated satellite-terrestrial network (ISTN) will share the spectrum. An efficient spectrum sharing mechanism is required to solve the problem of co-frequency interference while achieving seamless coverage and on-demand services. Existing studies have explored various approaches, including time-division and beam-hopping-based schemes, but the critical challenge of satellite-terrestrial time synchronization remains insufficiently addressed. This letter introduces an optimized time scheduling and alignment strategy based on time-division spectrum sharing between beam-hopping satellites and terrestrial cells. Specifically, a coordination unit plans the working time of multiple satellites and terrestrial cells according to the ephemeris, and further reduces the time synchronization cost through special timeslot configuration. Simulation results demonstrate the feasibility of the architecture and the improvement of communication capacity and time utilization.

Index Terms—Integrated satellite-terrestrial network, spectrum sharing, time-division, beam hopping, time synchronization.

I. INTRODUCTION

IN THE future 6G era, integrated satellite-terrestrial networks (ISTN) can provide ubiquitous connections for users with unified access devices, as promoted by the 3rd Generation Partnership Project (3GPP) [1]. However, the scarcity of spectrum resources requires spectrum sharing among these heterogeneous systems. Without proper coordination, co-frequency interference can severely degrade service quality [2], so it is necessary to coordinate spectrum utilization within ISTN to improve spectrum and network efficiency [3].

Traditional satellite and terrestrial communication networks adopt independent and fixed frequency bands, which leads to low spectrum utilization efficiency and poor compatibility. Later, dynamic spectrum sharing (DSS) schemes based

on cognitive radio were proposed to improve spectrum utilization by detecting available spectrum and adjusting the allocation [4]. However, the additional guard bands, sensing overhead, and the dynamic Doppler compensation requirements in non-geostationary orbit (NGSO) satellite systems make it quite challenging in practice.

With the improving accuracy of timing and timeslot allocation, it is feasible and more flexible to share spectrum in ISTN in a time-division manner. The key issue is how to handle the time-domain interference caused by the difference in satellite-terrestrial time delay and cell size. In terrestrial mobile communication, LTE-advanced (LTE-A) introduced the time-domain Enhanced Inter-cell Interference Coordination (eICIC) technology to solve the interference between heterogeneous cells, where macrocell base stations (BS) configure almost blank subframes (ABS) to provide communication opportunities for disturbed microcell users [5]. Some studies have explored the application of ABS and time-domain spectrum sharing in ISTN [6], [7], but they have not considered the time offset and corresponding interference caused by the movement of NGSO satellites.

In recent years, beam-hopping (BH) technology has been increasingly adopted by multi-beam satellite systems to enhance the flexibility of resource allocation [8]. BH naturally provides an opportunity for dynamic time-division spectrum sharing between satellite and terrestrial cells that are in the same space. However, existing research [9] still implicitly assumes the condition of geostationary satellites and thus lacks an integrated time allocation and synchronization mechanism suitable for the NGSO satellite systems.

Therefore, this letter introduces a dynamic time alignment and configuration strategy for time-division satellite-terrestrial spectrum sharing architecture, addressing a critical gap in ISTN design. The contributions of this letter are summarized as follows:

- A mechanism for adaptive time scheduling and alignment of multiple satellites and terrestrial cells is proposed for integrated beam-hopping NGSO satellites and terrestrial cellular networks.
- A special timeslot configuration method is proposed, which improves time utilization while maintaining low update overhead.
- Performance comparison with existing spectrum sharing and time synchronization strategies shows that it outperforms in terms of communication capacity and efficiency.

The remainder of this letter is organized as follows: Section II describes the adopted beam-hopping satellite-terrestrial spectrum sharing (BHSS) architecture, and

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Kai Chang is with the Department of Electronic Engineering, Tsinghua University, Beijing 100084, China (e-mail: changk22@mails.tsinghua.edu.cn).

Jianxiu Wang is with the 6G Research Center, China Telecom Research Institute, Beijing 102209, China (e-mail: wangjianxiu@chinatelecom.cn).

Bingkun Liu is with Beijing National Research Center for Information Science and Technology, Tsinghua University, Beijing 100084, China (e-mail: lbk@mail.tsinghua.edu.cn).

Linling Kuang is with Beijing National Research Center for Information Science and Technology and the State Key Laboratory of Space Network and Communications, Tsinghua University, Beijing 100084, China (e-mail: kll@tsinghua.edu.cn).

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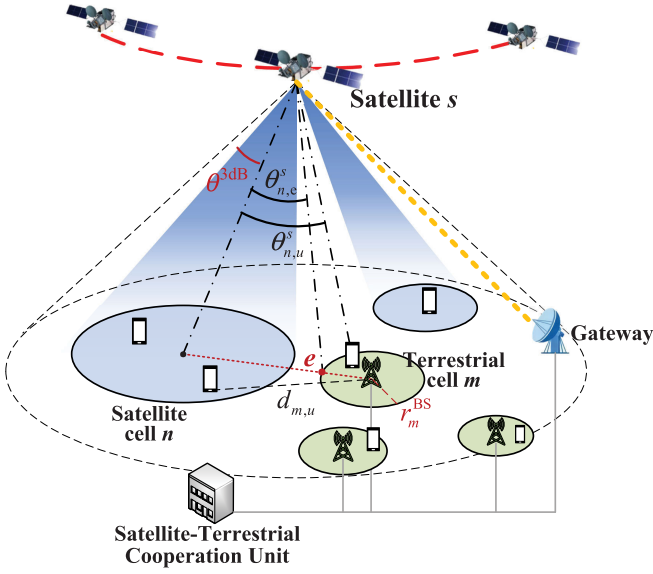


Fig. 1. The architecture of time-division satellite-terrestrial spectrum sharing with beam-hopping coverage.

Section III details the satellite-terrestrial time alignment strategy, including the special timeslot configuration. Simulation results and performance analysis are demonstrated in Section IV, followed by conclusions in Section V.

II. BEAM-HOPPING SPECTRUM SHARING ARCHITECTURE

The BHSS architecture typically consists of a NGSO satellite constellation and a terrestrial cellular network that contain common coverage area. Suppose there are a total of S visible satellites, N satellite cells and M terrestrial cells in the area we are concerned about, and the respective sets are denoted as $\mathcal{S}$, $\mathcal{N}$ and $\mathcal{M}$. It is assumed in our scenario that the satellite and terrestrial network share the same frequency band on the link in the same direction, and the following analysis is carried out from the perspective of downlink.

In BH, the satellite beams switch and point at the center of allocated satellite cells. To achieve seamless coverage in the marginal areas of the terrestrial network, the hopping satellite beams occasionally overlap with the terrestrial cells as shown in Fig. 1. In BHSS, the terrestrial BSs suspend downlink transmission when illuminated by satellite beams to avoid interference, while at other times they can resume the use of entire frequency band, realizing time-division spectrum sharing at each local space.

To obtain the time arrangement in BHSS ISTN, the beam-hopping time plan (BHTP) is scheduled first, and the terrestrial cells then decide the corresponding working time. A BHTP, which specifies the orientation of each satellite beam in a scheduling cycle, generally consists of $W \cdot L$ elements, meaning that W BH patterns are repeated for L times. The W patterns of satellite s can be denoted as

$$\begin{aligned} \mathbf{X}_s &= \{\mathbf{x}_s^1, \mathbf{x}_s^2, \dots, \mathbf{x}_s^t, \dots, \mathbf{x}_s^W\}, s \in \mathcal{S}, \\ \mathbf{x}_s^t &= \{x_{1,s}^t, x_{2,s}^t, \dots, x_{k,s}^t, \dots, x_{K_t,s}^t\}, x_{k,s}^t \in \mathcal{N}. \end{aligned} \quad (1)$$

$\mathbf{x}_s^t$ represents the BH pattern at timeslot t , and its element $x_{k,s}^t$, not identical to each other, indicates which satellite cell beam k is pointing at.

In order to coordinate the satellite and terrestrial network, we deploy a Satellite-Terrestrial Coordination Unit (STCU) to connect the cellular BSs and the satellite gateway stations. For each scheduling cycle, the STCU typically needs to carry out the following steps:

- BHTP scheduling: The STCU collects the distributed user demand and channel states and conducts the BH planning.
- Time arrangement calculation: Based on the BHTP result and the various satellite-terrestrial time delay, the accurate available working time of the terrestrial cells is calculated, which will be elaborated in the next section.
- Result configuration: The BHTP and terrestrial cell available time are assigned to each satellite and BS through the feeder link for the next cycle of operation.

To maximize the utilization of radio resources, it is necessary to clarify the satellite-terrestrial spacial interference relationship at each timeslot, as shown in Fig. 1. Suppose at a certain moment, one beam of satellite s points to satellite cell n , and the satellite beam width is θ^{3dB} . We use the binary variable $O_{n,m}^s$ to represent whether the satellite beam interferes with the terrestrial cell m . It is calculated by determining whether no terrestrial user of cell m is within the coverage range of the beam, and no satellite user served by the beam is within the range of cell m .

$$O_{n,m}^s = 1 - \prod_{u \in \mathcal{U}_m^{\text{BS}}} \delta\left(\theta_{n,u}^s > \frac{\theta^{3dB}}{2}\right) \prod_{u \in \mathcal{U}_{s,n}^{\text{Sat}}} \delta(d_{m,u} > r_m^{\text{BS}}), \quad (2)$$

where $\delta(x)$ is the indicator function, which means that if x is true, $\delta(x) = 1$; otherwise, $\delta(x) = 0$. $\mathcal{U}_m^{\text{BS}}$ and $\mathcal{U}_{s,n}^{\text{Sat}}$ respectively represent the sets of all users accessing BS m , and accessing satellite s while in the range of cell n . $\theta_{n,u}^s$ represents the angle between the line connecting user u and satellite s and the central axis of the beam, and $d_{m,u}$ represents the distance from user u to the BS. Assume that the BS is located at the center of the terrestrial cell, and r_m^{BS} represents the cell radius.

When the number of users is large, it is advisable to determine the interference relationship by judging whether there is an overlap between the satellite beam projection and the terrestrial cell. An approximate way is to check whether the intersection point e of the terrestrial cell edge and the line connecting the center of the terrestrial and satellite cell is within the coverage range of the satellite beam:

$$O_{n,m}^s = \delta\left(\theta_{n,e}^s \leq \frac{\theta^{3dB}}{2}\right). \quad (3)$$

The precise position of point e can be calculated by spherical interpolation. When using planar approximation, $\mathbf{p}_e = \frac{r_m^{\text{BS}}}{d_{n,m}} \mathbf{p}_n + \left(1 - \frac{r_m^{\text{BS}}}{d_{n,m}}\right) \mathbf{p}_m^{\text{BS}}$, where $\mathbf{p}_n$ and $\mathbf{p}_m^{\text{BS}}$ are the locations of the satellite and terrestrial cell center.

III. SATELLITE-TERRESTRIAL TIME SCHEDULING AND ALIGNMENT

This section will elaborate the optimized satellite-terrestrial time scheduling and alignment procedures we propose, as illustrated in Fig. 2. The target is to obtain the usable working time for each terrestrial cell at its local timeline,

based on the ephemeris and BHTP, etc.. As in BHTP, for each scheduling cycle, we deal with the first W timeslots and repeat the results for L times. The local time of the satellites and terrestrial BSs is synchronized with a uniform Global Navigation Satellite System (GNSS) time reference in advance.

According to the spacial interference relationship above-mentioned, the set of all satellite cells that interfere with the terrestrial cell m when served by satellite s is

$$\mathcal{N}_{s,m} = \{n \mid n \in \mathcal{N}, O_{n,m}^s = 1\}. \quad (4)$$

Then, we can generate the sequence of which satellites are going to interfere with the terrestrial cell m at each timeslot:

$$\begin{aligned} \mathbf{Y}_m &= \{\mathbf{y}_m^1, \mathbf{y}_m^2, \dots, \mathbf{y}_m^t, \dots, \mathbf{y}_m^W\}, \\ \mathbf{y}_m^t &= \{s \mid s \in \mathcal{S}, \exists n \in \mathcal{N}, n \in \mathbf{x}_s^t, n \in \mathcal{N}_{s,m}\}. \end{aligned} \quad (5)$$

$\mathbf{y}_m^t$ is the aggregation of the satellites that will serve the satellite cells that interfere with the terrestrial cell m at timeslot t according to the BHTP. If no corresponding satellite cells are going to be served at timeslot t , $\mathbf{y}_m^t$ is empty.

For a satellite s , its position $\mathbf{p}_{s,t}^{\text{Sat}}$ at timeslot t can be calculated based on the ephemeris. The satellite-terrestrial time delay at this time, which is the propagation time from the satellite to the center of the terrestrial cell, is

$$\tau_{s,m}^t = \frac{\|\mathbf{p}_{s,t}^{\text{Sat}} - \mathbf{p}_m^{\text{BS}}\|_2}{c}. \quad (6)$$

Hence, when a certain beam of the satellite illuminates the terrestrial cell m at timeslot t , the actual interference period at the BS's local timeline is $[(t-1) \cdot T + \tau_{s,m}^t, t \cdot T + \tau_{s,m}^t]$, where T is the timeslot length. Considering that the BHTP would be repeated L times, the satellite will illuminate the same cells at timeslot $t + l \cdot W$, where $0 \leq l \leq L-1, l \in \mathbb{N}$.

Since the satellite-terrestrial time delay varies among different repetitions due to the satellite movement, it is necessary to expand the interference period above. The maximum time delay $\tau_{s,m}^{\max}$ and minimum time delay $\tau_{s,m}^{\min}$ within the scheduling cycle can be obtained according to the ephemeris. Thereby, for timeslot t in the first W timeslots, the actual interference time period at the terrestrial BS within the scheduling cycle will not exceed

$$\xi_{s,m}^t = [(t-1) \cdot T + \tau_{s,m}^{\min}, t \cdot T + \tau_{s,m}^{\max}]. \quad (7)$$

As a result, the entire interference time of the terrestrial BS m can be obtained by taking the union of all the satellites s in $\mathbf{y}_m^t$ and all the timeslots from 1 to W :

$$\Xi_m^t = \bigcup_{s \in \mathbf{y}_m^t} \xi_{s,m}^t, \quad \Xi_m = \bigcup_{t \in \mathcal{W}} \Xi_m^t. \quad (8)$$

For each period of time in Ξ_m , it is evident that it can be expressed in the form of $\Xi_{m,i} = [(t_1-1) \cdot T + \tau_{s_1,m}^{\min}, t_2 \cdot T + \tau_{s_2,m}^{\max}]$, which refers to a continuous period affected by satellite signals from timeslot t_1 to t_2 . In order to separate the working time periods of the satellite and the terrestrial cell to cover the errors in ephemeris and clocks, etc., the guard period (GP) is required. The types of GP can be divided as the one between the satellite and the BS's working time GP_{s2c} , and the one between the BS and the satellite's working time GP_{c2s} . The GP is configured according to the system

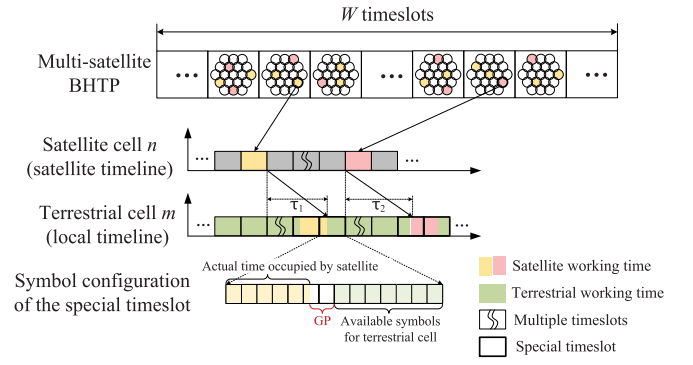


Fig. 2. An illustration of the timeslot synchronization and configuration mechanism.

conditions, but it can be noted that $GP_{c2s} \geq 2 \frac{r_m^{\text{BS}}}{c}$ due to the BS signal propagation within the cell range. We add GP to the unavailable time period of the terrestrial cell, since the satellite beams usually remain in one cell only for a short period of time. Therefore, the aforementioned non-transmissible period of the BS is extended to be $\Xi_{m,i} = [(t_1-1) \cdot T + \tau_{s_1,m}^{\min} - GP_{c2s}, t_2 \cdot T + \tau_{s_2,m}^{\max} + GP_{s2c}]$.

Next, we form the configuration of each timeslot for the terrestrial cell. We use $q_{ts}(\tau)$ and $r_{ts}(\tau)$ to represent the quotient and remainder of the time length divided by the timeslot length:

$$q_{ts}(\tau_{s_1,m}^{\min}) = \left\lfloor \frac{\tau_{s_1,m}^{\min}}{T} \right\rfloor, \quad r_{ts}(\tau_{s_1,m}^{\min}) = \tau_{s_1,m}^{\min} - q_{ts}(\tau_{s_1,m}^{\min}) \cdot T. \quad (9)$$

The preceding time period can thus be expressed as

$$\begin{aligned} &[(t_1 + q_{ts}(\tau_{s_1,m}^{\min} - GP_{c2s}) - 1) \cdot T + r_{ts}(\tau_{s_1,m}^{\min} - GP_{c2s}), \\ &(t_2 + q_{ts}(\tau_{s_2,m}^{\max} + GP_{s2c})) \cdot T + r_{ts}(\tau_{s_2,m}^{\max} + GP_{s2c})], \end{aligned}$$

which means on the local timeline of BS m , timeslots from $t_1 + q_{ts}(\tau_{s_1,m}^{\min} - GP_{c2s}) + 1$ to $t_2 + q_{ts}(\tau_{s_2,m}^{\max} + GP_{s2c}) - 1$ are fully unusable, while timeslots $t_1 + q_{ts}(\tau_{s_1,m}^{\min} - GP_{c2s})$ and $t_2 + q_{ts}(\tau_{s_2,m}^{\max} + GP_{s2c})$ are partially usable. We use matrix $\mathbf{A}_{M \times W}$ to record the state result of each timeslot of each terrestrial cell, and the values 0, 1, or 2 of its element a_{mt} respectively indicate that the timeslot t of cell m is unavailable, available, or partially available.

In order to make full use of the partially available timeslots, we introduce special timeslots as in the time division duplex (TDD) system. The criteria are that the first $ST_{s_1,m}^{c2s}$ symbols of the special timeslot between the working time period of the terrestrial cell and the satellite can be used by the BS, and so as the last $ST_{s_2,m}^{s2c}$ symbols of the special timeslot between the satellite and the terrestrial cell's working time. These two parameters can be calculated as

$$\begin{aligned} ST_{s_1,m}^{c2s} &= \left\lfloor \frac{r_{ts}(\tau_{s_1,m}^{\min} - GP_{c2s})}{T_{\text{sym}}} \right\rfloor, \\ ST_{s_2,m}^{s2c} &= N_{\text{sym}} - \left\lfloor \frac{r_{ts}(\tau_{s_2,m}^{\max} + GP_{s2c})}{T_{\text{sym}}} \right\rfloor. \end{aligned} \quad (10)$$

T_{sym} is the symbol length, and N_{sym} is the number of symbols in one timeslot. The summary of all the special

timeslot configurations of the terrestrial cells is aggregated as $ST_{M \times W}$, and the element is set as 0 if not a special timeslot. Moreover, if a timeslot is configured as two different types of timeslot, it should be set as unavailable. In this way, the configuration of each timeslot is determined. The entire process is summarized as Alg. 1.

Algorithm 1 BHSS Time Alignment and Configuration

Input: Locations of the satellite and terrestrial cells and users, the satellite constellation ephemeris and the multi-satellite BHTP result of the next scheduling cycle.

- 1: Initialize timeslot state matrix $A_{M \times W} = 1$.
- 2: **for** each terrestrial cell m in $\mathcal{M}$ **do**
- 3: Update the interference relationship O_{nm} and the disturbed satellite cell set $N_{I,m}$ based on the users' locations.
- 4: Generate the set of interfering satellites for each timeslot Y_m based on $\{X_s\}$ and $N_{I,m}$.
- 5: **for** each satellite s in $\mathcal{S}$ **do**
- 6: Calculate the maximum and minimum satellite-terrestrial delay $\tau_{s,m}^{\max}$ and $\tau_{s,m}^{\min}$ in the scheduling cycle, and the corresponding special timeslot configuration $ST_{s,m}^{s2c}$ and $ST_{s,m}^{c2s}$.
- 7: **end for**
- 8: Aggregate the occupied time of all satellites and all timeslots on the timeline of BS m to get Ξ_m .
- 9: **for** each time period in Ξ_m **do**
- 10: Set timeslot state a_{mi} from $t_1 + q_{ts}(\tau_{s1,m}^{\min} - GP_{c2s}) + 1$ to $t_2 + q_{ts}(\tau_{s2,m}^{\max} + GP_{s2c}) - 1$ as 0 (unavailable), and $t_1 + q_{ts}(\tau_{s1,m}^{\min} - GP_{c2s})$ and $t_2 + q_{ts}(\tau_{s2,m}^{\max} + GP_{s2c})$ as 2 (special timeslot). The special timeslot configurations are respectively $ST_{s,m}^{s2c}$ and $ST_{s,m}^{c2s}$.
- 11: The timeslot configured as a second type of special timeslot is set as unavailable.
- 12: **end for**
- 13: **end for**

Output: Timeslot state matrix $A_{M \times W}$ and special timeslot configuration matrix $ST_{M \times W}$ for all terrestrial cells.

IV. SIMULATION RESULTS AND PERFORMANCE ANALYSIS

A. Simulation Parameter

In this section, the proposed time scheduling and alignment method in BHSS architecture is simulated and verified. A schematic map is shown in Fig. 3, and simulation parameters are listed in Table I. The satellite orbit and communication capabilities refer to the Starlink, with an altitude of 550km, 48 orbital planes, 13 satellites in each orbital plane and an orbital inclination of 53° . The locations of terrestrial cells are derived from the data of existing LTE/NR networks, while the detailed transmission and reception capabilities of satellites, BSs and terminals can refer to [7] and [10]. All cells are set as a uniform size for simplicity. During the simulation period, there were a total of 4 visible satellites in the shown area. GP_{s2c} and GP_{c2s} are set respectively as 5us and 15us.

In order to simulate the non-uniform and time varying traffic, users are generated at random positions with a fixed

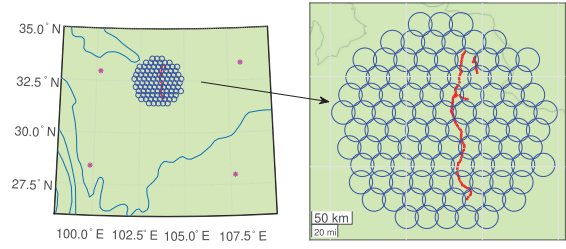


Fig. 3. A schematic map of the satellite cells (blue), terrestrial cells (red) and sub-satellite points (magenta) in the simulation area.

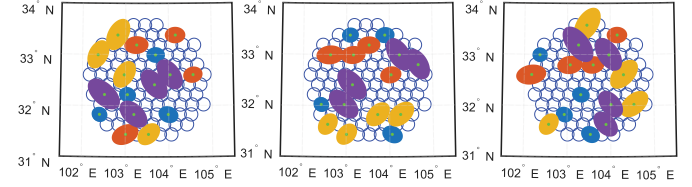


Fig. 4. An example of hopping beam projections of four satellites in three consecutive timeslots.

TABLE I
SIMULATION PARAMETERS

PARAMETERS	VALUES
Center position coordinate	(103.5°E, 32.5°N)
Satellite altitude	550 km
Satellite beam width θ^{3dB}	3°
Number of satellite cells N	100
Satellite cell radius	14.4 km
Number of terrestrial cells M	100
Terrestrial cell radius	1 km
Maximum beam number	4
Scheduling cycle	15s
Number of BH patterns W	1000
Downlink frequency	2GHz
System bandwidth	20MHz

proportion for the satellite and terrestrial network (1:10), and the traffic demand is subject to Poisson distribution. Specifically, different terrestrial cells have different user densities based on the public data, and the Poisson arrival rate is 5Mbps for each user. Multi-satellite BH scheduling with spatial isolation is conducted, which refers to [10]. Fig. 4 depicts the multiple hopping beam projections of the four satellites over three consecutive timeslots, which clearly shows the deformation of beam projection and the isolation between serving cells.

B. Simulation Results

1) System Service Performance:

We compare the time-aligned BHSS with several other spectrum sharing schemes under different loads:

- Interfered transmission: With no interference avoidance, disturbed users may face transmission interruption.
- Fixed frequency division: The available frequency band is evenly allocated to the satellite and terrestrial systems.

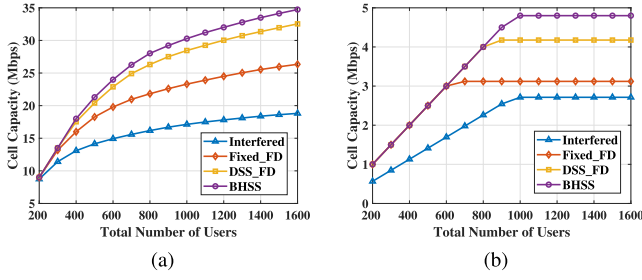


Fig. 5. The average service capacity for interfered (a) terrestrial cells and (b) satellite cells under different total numbers of users.

- DSS with frequency division: The usage of partial available spectrum is dynamically switched between satellite and terrestrial cells according to the load [4].

We use the average throughput rate of the disturbed cells over $L = 200$ scheduling cycles to represent the actual satellite and terrestrial service performance. In each scheduling period l , the throughput rate of cell i is the greater of the overall demand rate $D_{i,l}$ and the practical available capacity. Therefore, the average throughput rate of cell i can be expressed as

$$R_i = \text{avg}_{1 \leq l \leq L} \min \left(D_{i,l}, \frac{1}{W} \sum_{1 \leq t \leq W} \gamma_{i,t,l} C_{i,l} \right), \quad (11)$$

where $C_{i,l}$ is the theoretical capacity, and $\gamma_{i,t,l}$ is the actual working proportion of each timeslot. In this work, the γ of satellite cell is equivalent to whether it is served at this timeslot, and the γ of terrestrial cell can be obtained from the time scheduling result above.

As shown in Fig. 5, the performance is the worst when no interference avoidance is done. Under fixed frequency division, both satellite and terrestrial transmissions can only use half of the maximum bandwidth, thus the capacity upper limit is relatively low. The DSS method can dynamically allocate resources, while considering that not all the spectrum is shared and the overhead of guard bands, BHSS can increase the throughput of terrestrial cells by up to 7% and that of satellite cells by up to 15%.

2) Time Synchronization Efficiency:

We compare the proposed time alignment and configuration method in BHSS with several other methods to verify the time synchronization efficiency, which is measured by dividing the actual transmission capacity (deducting synchronization overhead) by the ideal allocated capacity.

- General time synchronization: The terrestrial cells synchronize time before each interference period of the satellite signal.
- Timeslot-based time alignment: The BHSS time alignment is carried out in units of timeslot in stead of symbol.
- Terrestrial-prior method: The satellites reduce the transmission of several symbols to align with the timeline of the terrestrial cells.

As depicted in Fig. 6, for disturbed terrestrial cells, the timeslot-based method causes more time waste as the timeslot length increases, so as the general time synchronization which requires more frequent synchronization updates. On the contrary, for disturbed satellite cells, the terrestrial-prior method

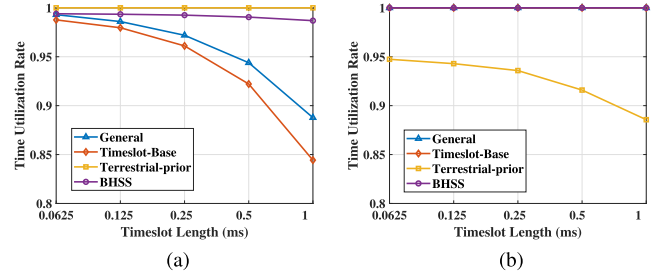


Fig. 6. The time synchronization efficiency of interfered (a) terrestrial cells and (b) satellite cells under different timeslot lengths.

brings great extra time loss, while the other methods can all utilize the whole time resource. Our proposed method can make preferable use of time resource in both cases, with the performance close to the ideal condition.

V. CONCLUSION

This letter establishes a viable time arrangement framework in BHSS that can effectively coordinate NGSO BH satellite and terrestrial cellular network. With the efficient time alignment and special timeslot configuration method, BHSS reaches high time utilization with low configuration overhead. Simulation analysis confirms the framework's effectiveness and advantage. This work offers a scalable solution for future ISTN in the 6G era, and lays a foundation for seamless spectrum sharing in dynamic environments.

REFERENCES

- [1] *Study on New Radio (NR) To Support Non-terrestrial Networks*, document TR 38.811, 3GPP, 2023.
- [2] A. Pastukh, O. Mironova, V. Tikhvinskiy, and V. Akhmediarov, "Analysis of the possibility to use hybrid satellite-terrestrial systems (Direct-to-device) in the IMT bands of terrestrial cellular networks," in *Proc. Syst. Signal Synchronization, Generating Process. Telecommun. (SYNCHROINFO)*, Jul. 2024, pp. 1–6.
- [3] J. Sun, X. Chen, C. Jiang, and S. Guo, "Distributionally robust optimization of on-orbit resource scheduling for remote sensing in space-air-ground integrated 6G networks," *IEEE J. Sel. Areas Commun.*, vol. 43, no. 1, pp. 382–395, Jan. 2025.
- [4] J. Zhu, Y. Sun, and M. Peng, "Beam management in low Earth orbit satellite communication with handover frequency control and satellite-terrestrial spectrum sharing," *IEEE Trans. Commun.*, vol. 73, no. 7, pp. 5247–5263, Jul. 2025.
- [5] S. Vasudevan, R. N. Pupala, and K. Sivasenan, "Dynamic eICIC — A proactive strategy for improving spectral efficiencies of heterogeneous LTE cellular networks by leveraging user mobility and traffic dynamics," *IEEE Trans. Wireless Commun.*, vol. 12, no. 10, pp. 4956–4969, Oct. 2013.
- [6] X. Wen, Y. Ruan, Y. Li, and R. Zhang, "Optimal cooperative transmission for overlay cognitive satellite terrestrial networks," *IEEE Commun. Lett.*, vol. 26, no. 2, pp. 419–423, Feb. 2022.
- [7] R. K. Saha, "Spectrum sharing in satellite-mobile multisystem using 3D in-building small cells for high spectral and energy efficiencies in 5G and beyond era," *IEEE Access*, vol. 7, pp. 43846–43868, 2019.
- [8] B. Liu, L. Kuang, and J. Lu, "Earth-fixed multicast user subgrouping for NGSO satellite with phased array antenna," *IEEE Internet Things J.*, vol. 11, no. 23, pp. 37661–37674, Dec. 2024.
- [9] M. Zhang, X. Yang, and Z. Bu, "Beam-hopping-based resource allocation in integrated satellite-terrestrial networks," *Sensors*, vol. 24, no. 14, p. 4699, Jul. 2024. [Online]. Available: <https://www.mdpi.com/1424-8220/24/14/4699>
- [10] Z. Lin, Z. Ni, L. Kuang, C. Jiang, and Z. Huang, "Satellite-terrestrial coordinated multi-satellite beam hopping scheduling based on multi-agent deep reinforcement learning," *IEEE Trans. Wireless Commun.*, vol. 23, no. 8, pp. 10091–10103, Aug. 2024.